VMAN FSDIVE PROBA

1 Motivation

DIVE example (fullscreen-switchable application). It (VMAN_Proba) is not the “best” example to show speed but (in hindsight) interesting because it lists all video modes. The latter functionality is in a separate utility program (gmis).

A port from Virtual Pascal to fpc Pascal (FPC 3.2.2).

2 VP code

2.1 Original sources (2002)

Original sources are published at Hobbes (?) and later at Github:

https://github.com/OS2World/DEV-SAMPLES-PASCAL-vman_fsdive

Original sources:

- main program
  - 25969 jun 13 2002 VMAN_Proba.pas
- supporting units:
  - 1590 mei 31 2002 FSDIVE.pas
  - 17671 mei 31 2002 GRADD.pas
  - 762 mei 31 2002 PMGRE.pas

2.2 compile with VP 2.1

Not a big issue (take care of compiler and linker switches).

3 FPC 3.2.2 units/packages

3.1 Intro

Typically C uses variables or pointers where Pascal will have variables and var + variable (Pascal can pass by value or by name, C cannot do that). Brute force replacing pointers by var leads to an issue when a nil pointer is allowed… Several changes that I propose share this background.

3.2 Unit mmbase

This defines constants and calculates them at end.

[20261004] mmioFourCC: in mmbase, but not exported:

```
Function mmioFOURCC(ch0,ch1,ch2,ch3:Char): Cardinal;
begin
  mmioFOURCC:=Cardinal(ord(ch0)) or (Cardinal(ord(ch1)) shl 8) or
        (Cardinal(ord(ch2)) shl 16) or (Cardinal(ord(ch3)) shl 24);
end;
```

3.3 Unit dive

[20261005] For dive.pas: 2 times boolean => longbool in record TDIVE_CAPS. [diveqcaps]

[20261102] more changes (from MAN_Proba):

- in Function DiveAllocImageBuffer: replace
  - Var bImageBuffer : Pointer
- by (remove var)
  - bImageBuffer : Pointer

[20261103] more changes (from game1src):

- in Functions DiveSetDestinationPalette, DiveSetSourcePalette: replace last arg
  - Var bRGB2Entries : Pointer)
- by (remove var)
  - bRGB2Entries : Pointer)

3.4 Unit pmgpi

Some changes: GpiQueryRegionRects, GpiCreateRegion:

GpiQueryRegionRects: change last arg

- var arclRectangles : RECTL) ;

into (remove var and change type)

- arclRectangles : pRECTL) ;

GpiCreateRegion: change third arg

- var prclBound : RECTL

into (remove var and change type)

- prclBound : pRECTL

4 FPC code

4.1 intro

Original code found on the www. This has a.o. units FSDIVE.PAS, GRADD.PAS, PMGRE.PAS.

4.2 function / what to expect

What you should get (ideally): window with contents that move up slowly. The pixel pattern and color is random and black/white shading varies gradually.

Special key: f: switches between window and full screen mode.

4.3 manual (how to run)

4.4 compile with fpc

Unresolved and fixed in unit divext.pas:

- **WinQueryVisibleRegion : pmwin.dll : import by index (1000)**
- **WinSetVisibleRegionNotify : pmwin.dll : import by index (1001)**

Found these (VP):

os2_os2_vp21/os2pmapi.pas: wm_VrnDisabled = $007E;

os2_os2_vp21/os2pmapi.pas: wm_VrnEnabled = $007F;

And so:

```
const
  wm_VrnDisabled  = $007E;
  wm_VrnEnabled  = $007F;
```

[20261005] From toolkit:

inc/bseord.inc: ORD_WIN32QUERYVISIBLEREGION EQU 1000

inc/bseord.inc: ORD_WIN32SETVISIBLEREGIONNOTIFY EQU 1001

From pmwin.h:

```
/*
** @86823; added WM_VRNDISABLED, WM_VRNENABLED,
**          WinSetVisibleRegionNotify, WinQueryVisibleRegion
*/
#define WM_VRNDISABLED          0x007e
#define WM_VRNENABLED           0x007f

BOOL APIENTRY WinSetVisibleRegionNotify( HWND hwnd,
                                         BOOL fEnable);

ULONG APIENTRY WinQueryVisibleRegion( HWND hwnd,
                                      HRGN hrgn);
```

4.5 Unit FSDIVE

Add cdecl to all functions.

{ we need cdecl here }

4.6 Unit GRADD

4.6.1 gradd.pas

[20261005] Remarks on GRADD for FPC:

uses os2def;

type BOOL=boolean;

// type ULONG=cardinal; {ok but no change}

{ we need cdecl here }

[20261101] From checking/comparing record sizes (VP vs fpc): 2 essential changes in GRADD.PAS:

- change longbool into boolean
- add {$PACKRECORDS 1}

4.6.2 demo: gmis.pas

Demo utility program: gmis (to get GRADD modes). Simplified as much as possible. Depends on (uses) VMAN.DLL and GRADD. Unknown what it does with non-GRADD but could be OK (think it misses the refresh rate).

4.7 Unit PMGRE

PMGRE fixed for FPC:

type HDC=cardinal;

type long=longint;

{ we need cdecl here }

4.8 Program VMAN_Proba

From dive: SETUP_BLITTER >Tsetup_blitter (dive)

os2_os2_vp21/os2mm.pas: Setup_Blitter = record

os2_os2_vp21/os2mm.pas: ulStructLen : ULong; // Set equal to sizeof(SETUP_Blitter)

os2_os2_vp21/os2mm.pas: pSetup_Blitter = ^Setup_Blitter;

os2_os2_vp21/os2mm.pas:Function DiveSetupBlitter(hDiveInst: hDive; pSetupBlitter: pSetup_Blitter): ULong;

From doscalls:

function DosLoadModule (ObjName: PChar; ObjLen: cardinal; DLLName: PChar;

var Handle: THandle): cardinal; cdecl;

function DosLoadModule (var ObjName: string; ObjLen: cardinal;

const DLLName: string; var Handle: THandle): cardinal;

function DosQueryProcAddr (Handle: THandle; Ordinal: cardinal; ProcName: PChar;

var Address: pointer): cardinal; cdecl;

Semaphore/mutex: if needed can use this:

var hmtxDiveMutex: cardinal;

[20261103] Overview of some non-trivial changes:

[1] Change (VP)

rc:=DosQueryProcAddr(hmodVMAN, 0, 'VMIEntry', pfn(@VMIEntry)); // Query entry point address

into (fpc)

rc:=DosQueryProcAddr(hmodVMAN, 0, 'VMIEntry', pointer(@VMIEntry)); // Query entry point address

OK.

[2] Change (VP)

move(mem[ulong(pbBuffer)+j*ulScanLineBytes], mem[ulong(pbBuffer)+(j+1)*ulScanLineBytes], ulScanLineBytes);

into (fpc) (psrce, pdest declared as pointers)

psrce:=pbBuffer+(j+0)*ulScanLineBytes;

pdest:=pbBuffer+(j+1)*ulScanLineBytes;

move(psrce^, pdest^, ulScanLineBytes);

4.9 Program gmis (utility)

Source code is in gmis.pas. This little program was quite useful in debugging the previous.

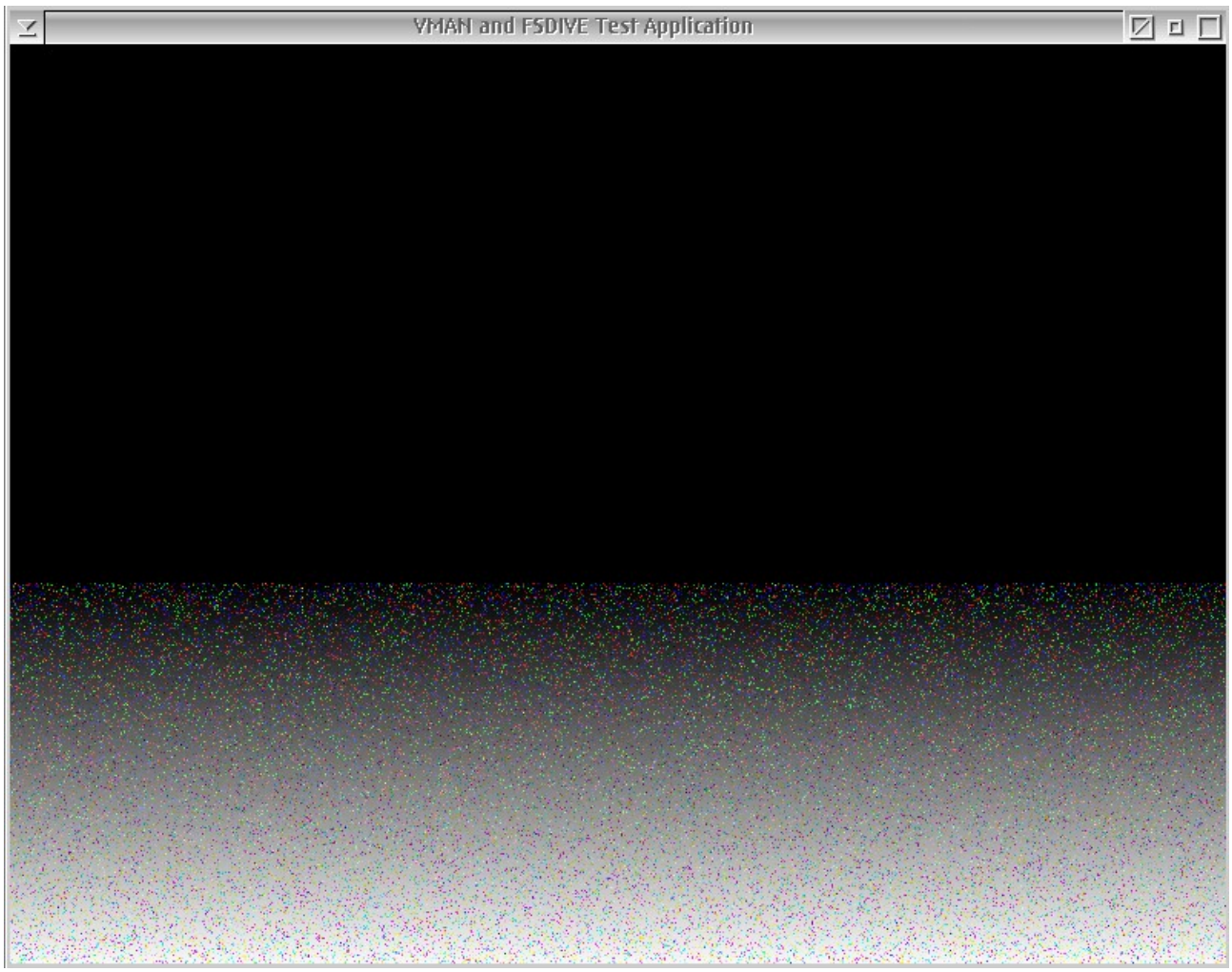
Functionality of gmis: it writes two files:

- gmis.log, with all the video modes found
- tryit.cmd, a script to run VMAN_Proba based on the found modes. Hit Alt-F4 to go to the next variation. Note that sometimes only a frame is drawn, and the frame can be larger than the display (!).

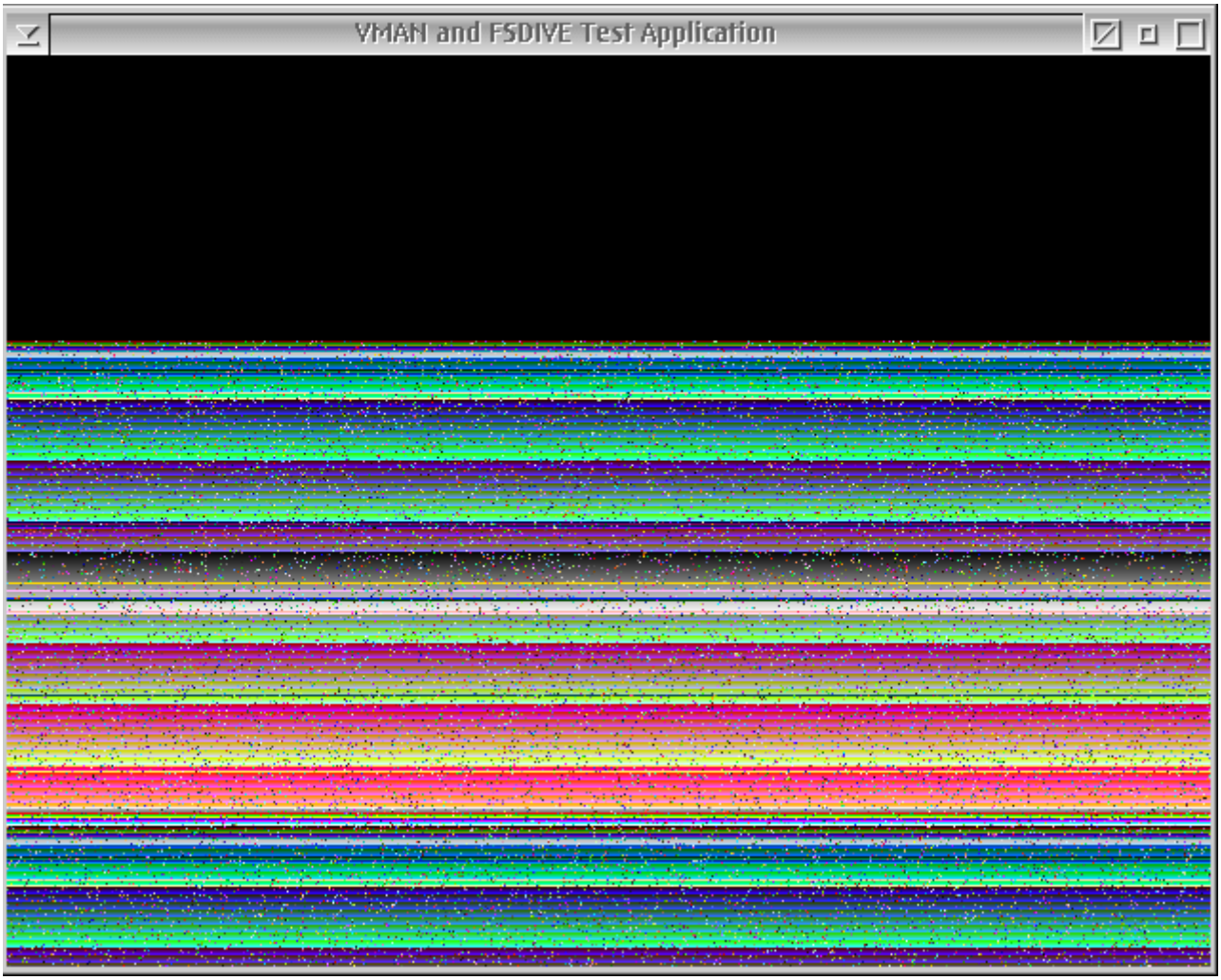
4.10 Running VMAN_Proba

See gmis section above. The f key switches between windowed and Full Screen modes. You can also use the standard window widget to maximize the window.

Examples of windowed output (ArcaOS):


VMAN and FSDIVE Test Application

Or (less nice):



4.11 Quality

Compilation ends with 3 warnings and 2 notes.

The program can be compiled with -Cr -gh; the result runs.

5 Testing

Ideally the program is tested at:

- hardware/metal
- VM

Display drivers like:

- GRADD
- Snap
- Panorama.

Relevant OS/2 versions:

- latest IBM (4.50 aka Warp 4.50)
- latest eCS (2.1)
- latest ArcaOS (5.1)

That gives 18 use cases or 12 when I omit eCS.

I did most of my testing with Warp 4.50 with GRADD and ArcaOS with Panorama. Surprisingly Panorama gave best results.

6 Revisions

[20261106] pub.

[20261111] pub. Was too fond of pointers. Shorter & simplified.